

Technical Report

Term Deposit Customer Ranking

Objective. Validate a pre-call ranking system for prioritising already eligible customers for term-deposit calls.

Evidence base. Final executed notebooks 01-05; 40,000 labelled historical calls; deterministic SEED=42.

40,000	7.24%	12
historical calls	subscription rate	deployable pre-call predictors

Technical conclusion

HGB is the stronger conventional shuffled-validation challenger, but Logistic Regression is stronger in ordered expanding-window validation and period-local lift. Use unweighted all-12 pre-call LR for a controlled pilot; retain HGB as challenger. Treat scores as ranking signals, not calibrated probabilities.

Executive Summary

The business task is capacity allocation, not a definitive yes/no classification at 0.50. The model should rank already eligible customers before a call so that limited calling capacity is concentrated near the top of the list.

Only 2,896 of 40,000 calls ended in subscription (7.24%). Accuracy is reported, but ranking metrics, ordered robustness, calibration diagnostics and call-depth lift are more informative for this imbalanced problem.

duration is created after the call and is therefore excluded from all deployable modelling. Its correlation with subscription is 0.461169 and adding it lifts retrospective ROC-AUC above 0.93, demonstrating the scale of post-call leakage.

On the untouched holdout, tuned HGB has higher pre-call PR-AUC/ROC-AUC (0.272820/0.733930) than tuned LR (0.240265/0.709681). In the stricter ordered evaluation the preference reverses: LR weighted PR-AUC/ROC-AUC is 0.112626/0.566290 versus HGB 0.096333/0.551600.

At 5% period-local call depth, LR captures 228 subscribers from 1,528 calls versus about 123 expected under random calling at the same volume: 1.853857x lift. LR lift remains 1.606389x at 10% depth and 1.401474x at 20% depth.

Decision

Proceed to controlled pilot planning with unweighted all-12 Logistic Regression. Keep HGB as challenger. Do not approve unrestricted production deployment until live stability, calibration and economics are measured.

1. Decision Context and Prediction Moment

Operational flow: eligible customers -> pre-call score -> ranked list -> call highest-ranked customers first -> stop when capacity is exhausted.

The model does not determine eligibility, approve customers, set prices or guarantee subscription. Any feature created during or after the call is invalid for this ranking decision.

1.1 Deployable predictors

The 12 valid pre-call predictors are age, job, marital, education, default, balance, housing, loan, contact, day, month and campaign. duration is excluded.

1.2 Why ranking metrics matter

PR-AUC and ROC-AUC evaluate score ordering across thresholds. Lift translates that ordering into a capacity question: how many subscribers are concentrated within the top-ranked fraction relative to random calling at the same volume?

Threshold-tuned F1 is a hard-classification diagnostic, not the operating policy.

2. Data Audit and Exploratory Evidence

The canonical analysis contains 40,000 historical calls. The helper `y_binary` is dropped if present and recreated from the original yes/no target in every notebook.

Table 1: Dataset and target audit

Measure	Result
Rows	40,000
Subscribers	2,896
Non-subscribers	37,104
Subscription rate	7.24%
Majority-class accuracy	92.76%
Exact duplicate source rows	0
Conventional null values	0
Blank strings	0

2.1 Explicit unknown categories

There are no conventional nulls, but literal unknown categories are retained as explicit levels rather than silently imputed.

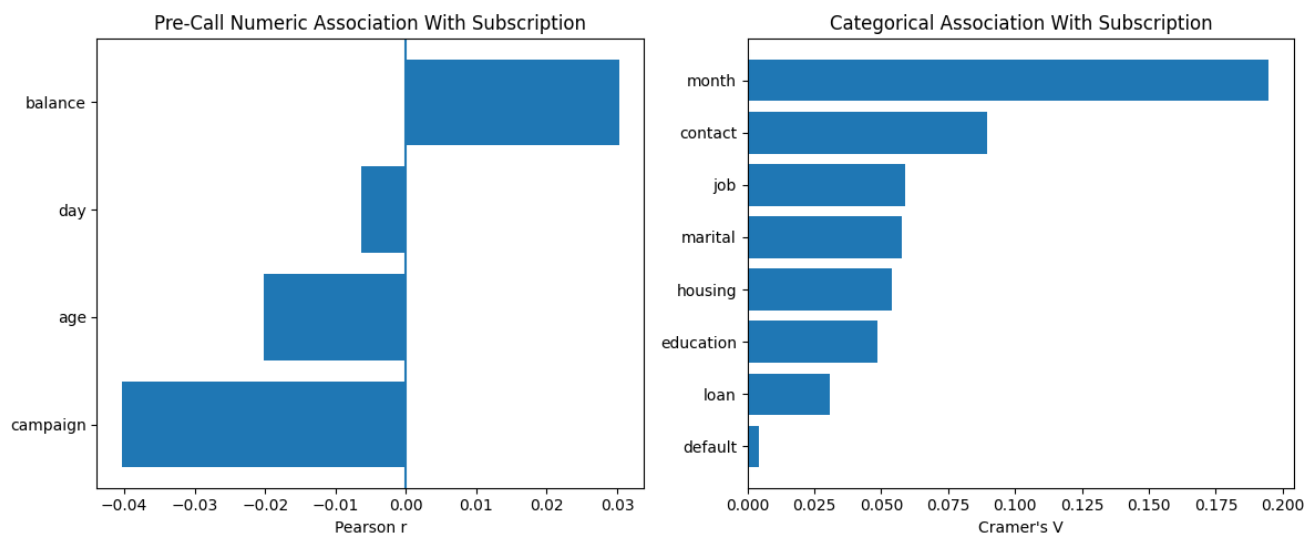
Table 2: Literal unknown categories

Feature	Records	Share
contact	12,765	31.91%
education	1,531	3.83%
job	235	0.59%

2.2 Pre-call associations

Pearson correlation is used for numeric pre-call variables and bias-corrected Cramer's V for categoricals. These are marginal associations, not causal effects or model importance.

Fig. 1: Pre-call numeric Pearson association and categorical Cramer's V association with subscription.

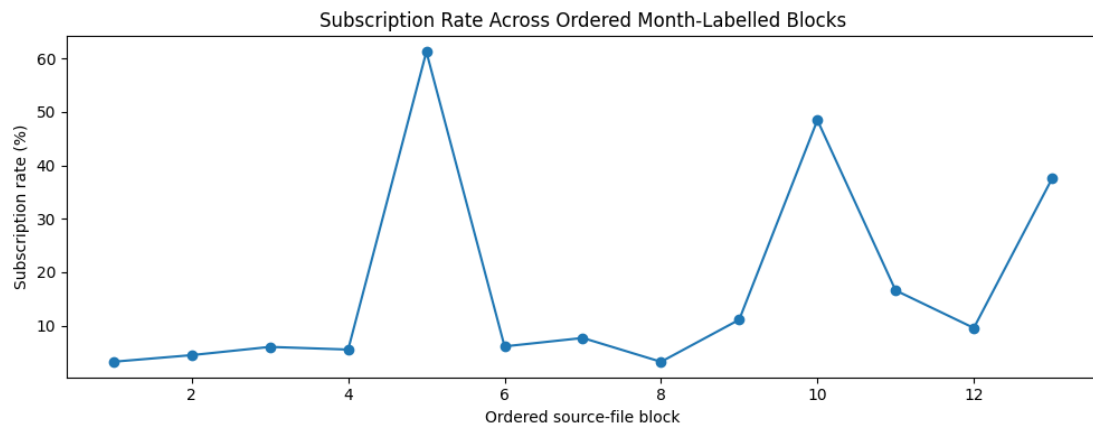


Month has the strongest categorical association (Cramer's V 0.19453), followed by contact (0.08958). Numeric associations are weak: campaign -0.04035, balance +0.03023, age -0.02027 and day -0.00642.

2.3 Ordered source blocks

Month labels have no verified year or timestamp. Consecutive changes form 13 source-order month-labelled blocks; these must not be described as confirmed calendar periods.

Fig. 2: Subscription Rate Across Ordered Month-Labelled Blocks.



Interpretation

The sharp variation in block-level subscription rate is an early warning that shuffled validation can overstate future-like stability.

3. Leakage Diagnostic: Call Duration

Call duration has Pearson $r = 0.461169$ with subscription and is known only after a call has taken place. The with-duration experiment therefore quantifies leakage; it is not a deployable feature test.

Table 3: Ranking impact of adding post-call duration

Stage	Model	Pre-call PR	+dur PR	PR change	Pre-call ROC	+dur ROC	ROC change
Default	LR	0.241676	0.525049	+0.283373	0.711156	0.932379	+0.221223
Default	HGB	0.268264	0.585190	+0.316926	0.734102	0.951528	+0.217425
Tuned	LR	0.240265	0.520693	+0.280427	0.709681	0.933188	+0.223507
Tuned	HGB	0.272820	0.549825	+0.277004	0.733930	0.943115	+0.209185

Leakage conclusion

duration is excluded from deployable feature importance, feature reduction, imbalance experiments, robustness validation, calibration and lift.

4. Model Families, Validation and Hyperparameter Selection

Logistic Regression provides a transparent linear baseline; HistGradientBoosting is a nonlinear challenger able to capture interactions and nonlinear structure. This is a focused linear-vs-nonlinear comparison, not an exhaustive classifier benchmark.

4.1 Deterministic split design

SEED=42 defines a 32,000-row development set and untouched 8,000-row final holdout. Development is split into 24,000 tuning-train and 8,000 tuning-validation rows. HGB search uses a reproducible 10,000-row stratified subsample. Hyperparameter selection is a fixed train/validation experiment, **not K-fold tuning CV**.

4.2 Search spaces and objective

LR evaluates 15 log-spaced C values from 0.001 to 100 x L1/L2 x none/balanced class weight: 60 configurations per feature condition. HGB defines 28,800 possible combinations over iterations 50-400, depth 3/4/5/6/8/None, learning rate 0.01-0.20, 15-127 leaves, L2 0-2, min leaf 10-50 and none/balanced class weight; ParameterSampler evaluates the same 100 candidates per feature condition with random_state=42. Across four conditions, exactly **320 candidate configurations** are evaluated. PR-AUC is primary; ROC-AUC breaks ties.

4.3 Selected configurations

Table 4: Fresh selected hyperparameters

Configuration	Selected parameters
LR pre-call	L2; C=8.4834289824; unweighted
LR + duration	L1; C=0.0610540230; unweighted
HGB pre-call	200 iterations; depth 8; LR 0.05; 15 leaves; L2=0; min leaf 20; unweighted; early stopping
HGB + duration	200 iterations; depth 3; LR 0.05; 127 leaves; L2=2; min leaf 10; balanced; early stopping

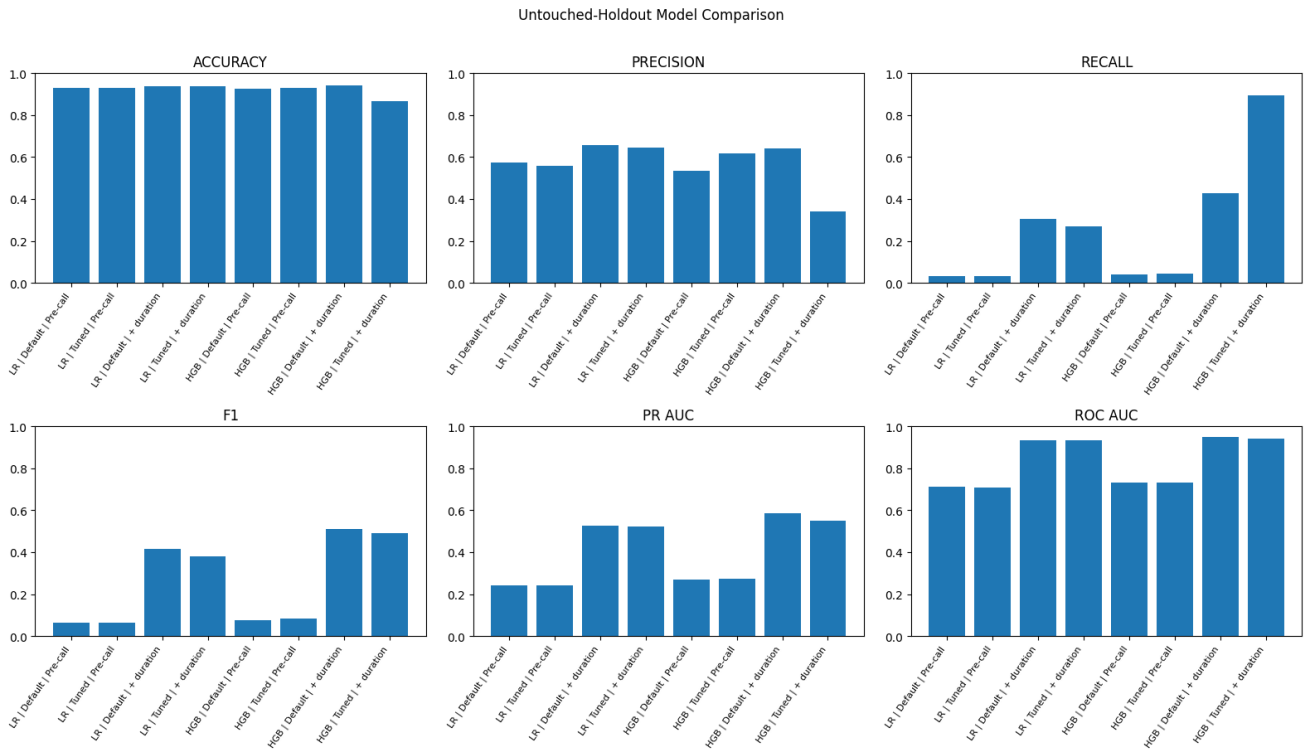
Because the splits, HGB subsample, ParameterSampler sequence and model randomness are seeded, rerunning Notebook 2 in the same environment reproduces the same tuning experiment and winners.

4.4 Untouched final-holdout results

Table 5: Default and tuned holdout performance

Stage	Model	Features	Accuracy	Precision	Recall	F1	PR-AUC	ROC-AUC
Default	LR	Pre-call	0.928250	0.575758	0.032815	0.062092	0.241676	0.711156
Tuned	LR	Pre-call	0.928125	0.558824	0.032815	0.061990	0.240265	0.709681
Default	HGB	Pre-call	0.928000	0.534884	0.039724	0.073955	0.268264	0.734102
Tuned	HGB	Pre-call	0.928875	0.619048	0.044905	0.083736	0.272820	0.733930
Default	LR	+duration	0.938125	0.655556	0.305699	0.416961	0.525049	0.932379
Tuned	LR	+duration	0.936500	0.646091	0.271157	0.381995	0.520693	0.933188
Default	HGB	+duration	0.941125	0.639896	0.426598	0.511917	0.585190	0.951528
Tuned	HGB	+duration	0.866000	0.338572	0.892919	0.490978	0.549825	0.943115

Fig. 3: Six-metric untouched-holdout comparison for default/tuned LR and HGB, with and without post-call duration.



Pre-call tuning gives almost no LR ranking improvement and only a small HGB PR-AUC gain. Hyperparameter optimisation does not remove the core robustness problem.

5. Feature Evidence and Reduction

Marginal Pearson/Cramer's V, LR coefficient magnitude and HGB permutation importance answer different questions and are not merged into a synthetic combined ranking.

5.1 Association versus fitted-model importance

Table 6: Association and model-importance comparison

Feature	Assoc.	LR [coef]	HGB perm.	Assoc rank	LR rank	HGB rank
month	0.194534	3.088867	0.168555	1	1	1
contact	0.089584	0.922315	0.119955	2	2	2
day	0.006420	0.106809	0.081196	11	10	3
housing	0.054106	0.571184	0.025086	5	3	4
age	0.020273	0.030198	0.011867	10	12	5
balance	0.030232	0.050513	0.005755	9	11	6
marital	0.057858	0.458118	0.003635	4	5	7
campaign	0.040350	0.159715	0.003570	7	9	8
loan	0.030624	0.484238	0.002746	8	4	9
education	0.048393	0.281802	0.002696	6	8	10
default	0.004246	0.433878	-0.000307	12	6	11
job	0.058897	0.384430	-0.000900	3	7	12

The disagreement is informative: day has weak marginal association but ranks third under HGB permutation importance, while job has stronger marginal association but near-zero/negative HGB permutation importance.

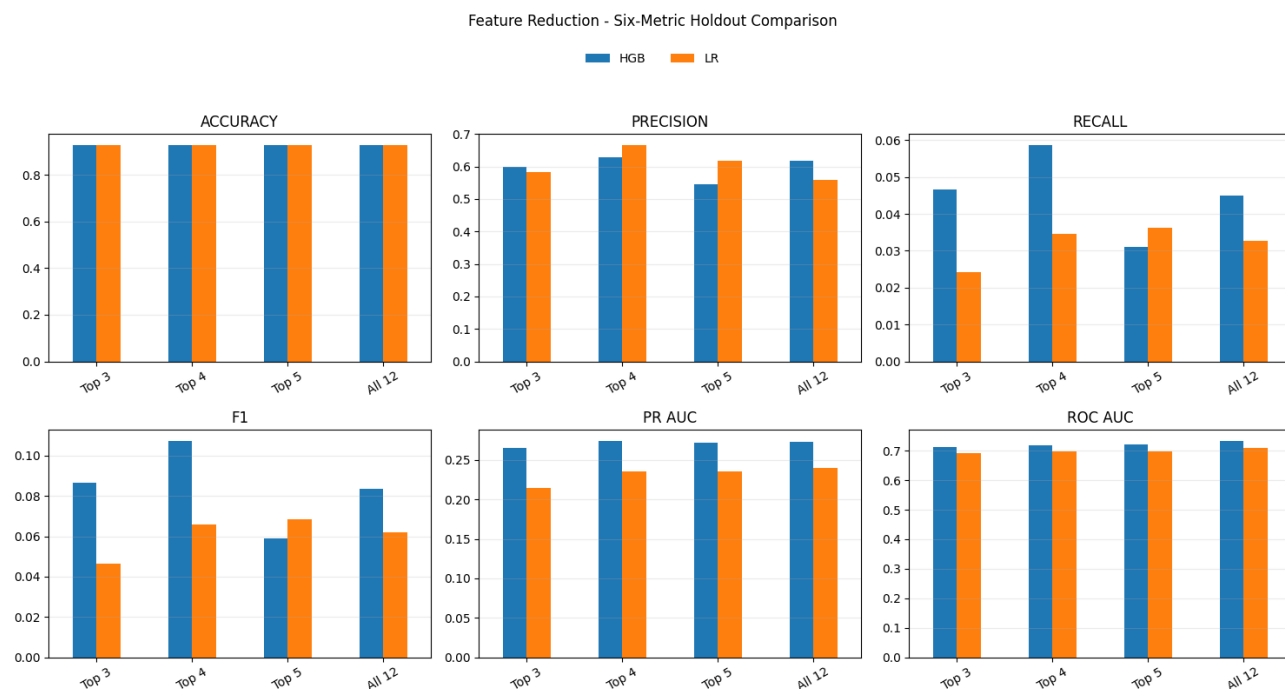
5.2 Feature-reduction experiment

HGB permutation importance alone defines the reduction order: Top-3 month/contact/day; Top-4 adds housing; Top-5 adds age. Reduced subsets reuse the same tuned model structures; there is no subset-specific retuning.

Table 7: Feature-reduction holdout metrics

Set	Model	Accuracy	Precision	Recall	F1	PR-AUC	ROC-AUC
Top 3	LR	0.928125	0.461538	0.024180	0.046434	0.214884	0.691444
Top 3	HGB	0.928750	0.562500	0.046632	0.086538	0.265716	0.713004
Top 4	LR	0.928875	0.612903	0.034542	0.065681	0.235485	0.698631
Top 4	HGB	0.929375	0.654545	0.058722	0.107425	0.274531	0.719196
Top 5	LR	0.928625	0.586207	0.036269	0.068515	0.235275	0.699099
Top 5	HGB	0.928000	0.500000	0.031088	0.058824	0.272162	0.720062
All 12	LR	0.928125	0.558824	0.032815	0.061990	0.240265	0.709681
All 12	HGB	0.928875	0.619048	0.044905	0.083736	0.272820	0.733930

Fig. 4: Six-metric feature-reduction comparison using fixed tuned pre-call structures.



HGB Top-4 has slightly higher PR-AUC than All-12 (0.274531 vs 0.272820), but All-12 has materially stronger ROC-AUC (0.733930 vs 0.719196). The pilot retains all 12 valid pre-call predictors.

6. Class Imbalance and Threshold Behaviour

Six strategies are compared on the tuned pre-call structures: none, balanced class weighting, random oversampling, random undersampling, SMOTENC and threshold tuning.

6.1 Design

Weighting/resampling changes training and can alter ranking. Threshold tuning instead uses five-fold out-of-fold development predictions to choose the F1-maximising cutoff from 0.005-0.995, then applies it once to the untouched holdout. It does not change the continuous scores, so PR-AUC and ROC-AUC are unchanged.

6.2 Results and operational interpretation

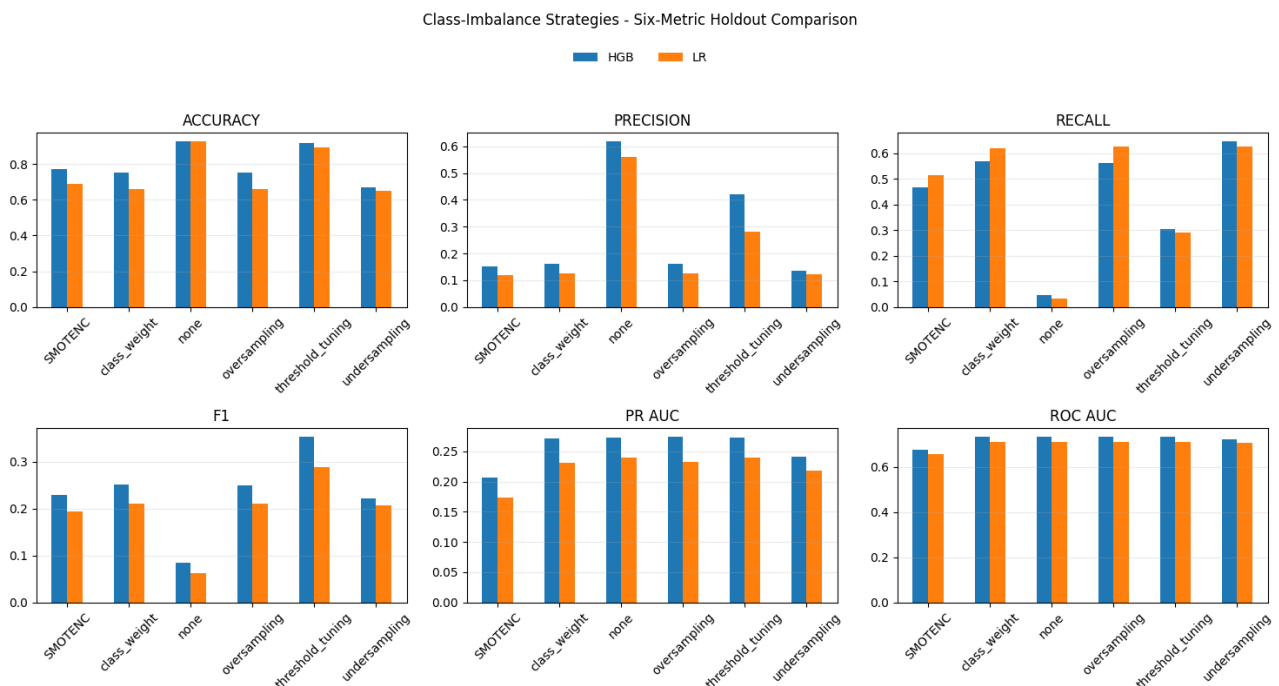
Table 8: Logistic Regression imbalance strategies

Strategy	Thr.	Accuracy	Precision	Recall	F1	PR-AUC	ROC-AUC
None	0.500	0.928125	0.558824	0.032815	0.061990	0.240265	0.709681
Class weight	0.500	0.662875	0.126587	0.620035	0.210249	0.231243	0.710575
Oversampling	0.500	0.661000	0.126967	0.626943	0.211169	0.232024	0.710790
Undersampling	0.500	0.651875	0.123549	0.625216	0.206327	0.218951	0.708137
SMOTENC	0.500	0.689875	0.119295	0.514680	0.193695	0.172603	0.656204
Threshold tuning	0.141	0.895125	0.282609	0.291883	0.287171	0.240265	0.709681

Table 9: HistGradientBoosting imbalance strategies

Strategy	Thr.	Accuracy	Precision	Recall	F1	PR-AUC	ROC-AUC
None	0.500	0.928875	0.619048	0.044905	0.083736	0.272820	0.733930
Class weight	0.500	0.754750	0.161527	0.569948	0.251716	0.271896	0.732946
Oversampling	0.500	0.753625	0.159491	0.563040	0.248570	0.274559	0.733041
Undersampling	0.500	0.671750	0.134072	0.647668	0.222156	0.241083	0.723593
SMOTENC	0.500	0.775250	0.150716	0.454231	0.226334	0.210334	0.674565
Threshold tuning	0.156	0.919375	0.421053	0.303972	0.353059	0.272820	0.733930

Fig. 5: Six-metric class-imbalance comparison for LR and HGB.



HGB oversampling raises PR-AUC by only 0.001739 while accuracy and precision fall sharply. SMOTENC worsens ranking. Threshold tuning produces the strongest F1 (LR 0.287171; HGB 0.353059) but not better ordering. The operational recommendation remains unweighted ranking with call depth chosen from capacity/economics.

7. Exploratory Higher-Conversion Population

The descriptive rule `contact=cellular AND (job=retired OR age>=71 OR balance>=Q3)` identifies 6,774 customers, 778 subscribers and 11.485% conversion versus 7.24% overall. This is exploratory context, not a demographic targeting rule.

For validation, Q3 is learned inside each training fold; 6,772 rows meet fold-local definitions. Default model structures are used with no subgroup-specific retuning.

Table 10: Exploratory population default-model validation

Model	Rows	Base rate	Accuracy	Precision	Recall	F1	PR-AUC	ROC-AUC	Brier
LR	6,772	0.114885	0.883934	0.420000	0.026992	0.050725	0.286349	0.692504	0.092816
HGB	6,772	0.114885	0.884968	0.497537	0.129820	0.205912	0.303476	0.694120	0.092533

Boundary on interpretation

This subgroup result is not causal, does not prove fairness, and does not justify replacing the ranking model with a demographic rule.

8. Robustness: Shuffled, Pseudo-Grouped and Ordered Validation

Three robustness views are kept separate: ordinary shuffled five-fold validation, pseudo-profile grouped sensitivity and expanding-window source-order evaluation.

8.1 Shuffled and pseudo-profile sensitivity

Table 11: Shuffled and pseudo-profile ranking metrics

Validation	Model	PR-AUC	ROC-AUC	Brier
Shuffled 5-fold	LR	0.215433	0.684767	0.062800
Shuffled 5-fold	HGB	0.245333	0.717461	0.060778
Pseudo-profile grouped	LR	0.215690	0.685003	0.062797
Pseudo-profile grouped	HGB	0.248093	0.716737	0.060719

Pseudo-groups use age+job+marital+education+balance. Similarity to shuffled results is reassuring, but there is no true customer ID, so this cannot prove customer-level independence.

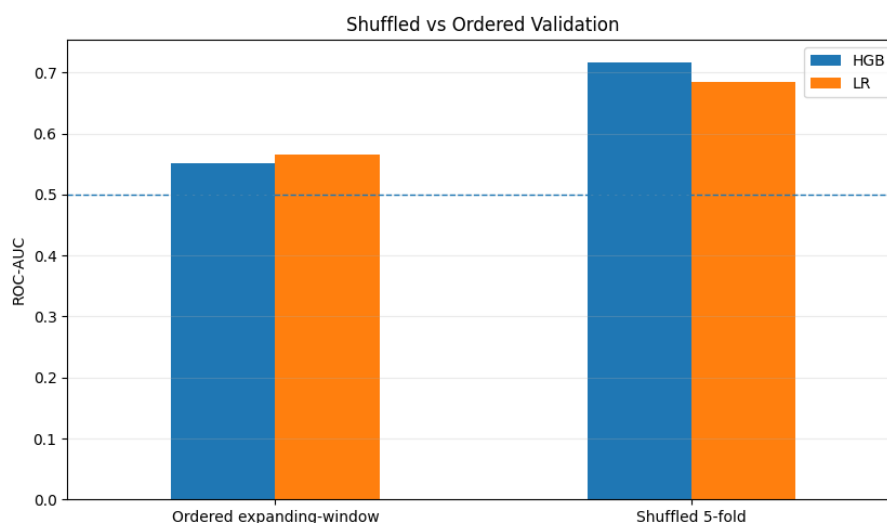
8.2 Ordered expanding-window evaluation

A block is eligible only with at least 5,000 earlier training rows and at least 100 positive and 100 negative test outcomes. Eight of 13 ordered blocks pass, covering 30,526 test rows and 2,457 positives.

Table 12: Ordered expanding-window summary

Model	Periods	Rows	Positives	Weighted PR-AUC	Weighted ROC-AUC	Weighted ROC SD	Weighted Brier
LR	8	30,526	2,457	0.112626	0.566290	0.040213	0.074776
HGB	8	30,526	2,457	0.096333	0.551600	0.023037	0.074181

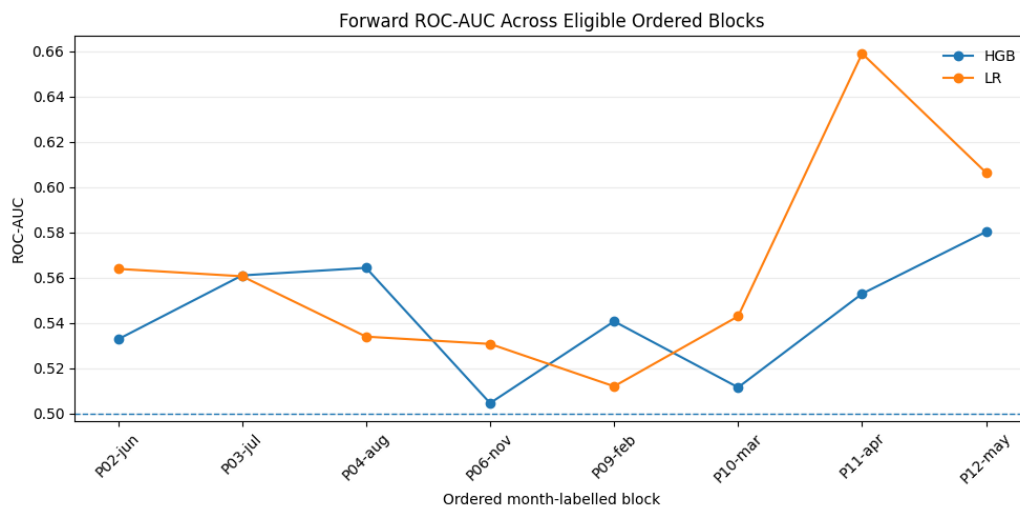
Fig. 6: Shuffled vs Ordered Validation.



Central robustness result

HGB wins shuffled validation; LR wins the stricter ordered test. Both degrade substantially. This is the main reason to recommend a controlled pilot rather than unrestricted deployment.

Fig. 7: Forward ROC-AUC Across Eligible Ordered Blocks.



Period-level discrimination varies sharply, with several blocks near chance and LR notably stronger in later eligible blocks. These are source-order month labels, not verified years.

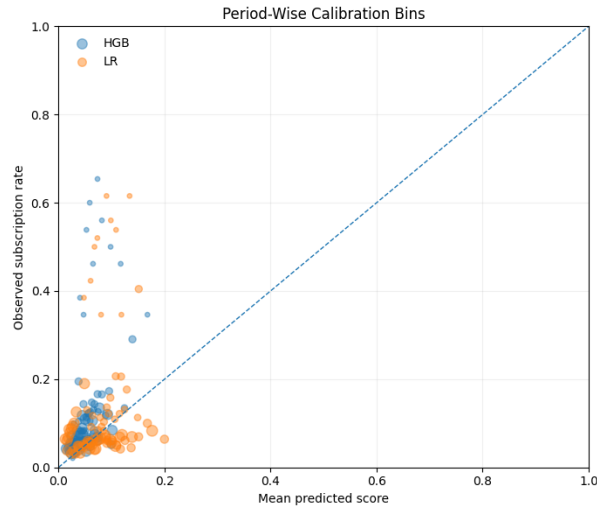
9. Calibration Diagnostics and Capacity-Based Ranking

The final notebook evaluates period-wise score bins, Brier score and expected calibration error without fitting sigmoid or isotonic recalibration. Ranking and calibration are therefore interpreted separately.

Table 13: Ordered calibration summary

Model	Periods	Rows	Base rate	Mean score	Brier	ECE	ROC-AUC	PR-AUC
LR	8	30,526	0.080489	0.075410	0.074776	0.044442	0.566290	0.112626
HGB	8	30,526	0.080489	0.048578	0.074181	0.035101	0.551600	0.096333

Fig. 8: Period-wise calibration bins for forward predictions. The diagonal denotes perfect calibration.



Raw outputs should be treated as ranking signals, not exact probabilities. Probability-based revenue calculations require prospective recalibration on new campaign data.

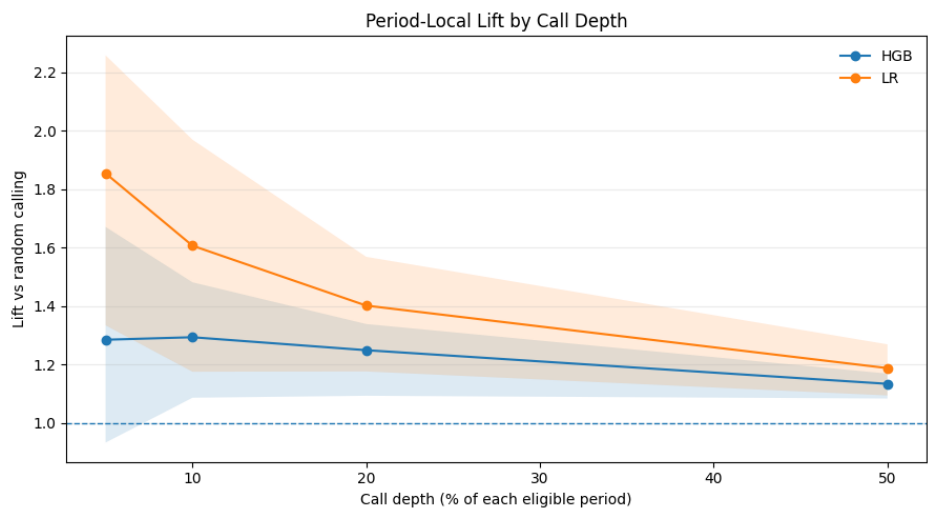
9.1 Period-local lift

Within each eligible block, customers are ranked and the top fraction is compared with random selection at the same volume. Counts are then aggregated across blocks.

Table 14: Period-local call-depth lift

Model	Depth	Called	Captured	Random exp.	Incremental	Lift	Capture rate	95% CI
LR	5%	1,528	228	122.99	105.01	1.853857	0.092796	1.333945-2.257328
LR	10%	3,055	395	245.89	149.11	1.606389	0.160765	1.175284-1.968856
LR	20%	6,108	689	491.63	197.37	1.401474	0.280423	1.176270-1.568156
LR	50%	15,264	1,459	1,228.58	230.42	1.187549	0.593814	1.094145-1.268835
HGB	5%	1,528	158	122.99	35.01	1.284690	0.064306	0.932990-1.670341
HGB	10%	3,055	318	245.89	72.11	1.293245	0.129426	1.086337-1.481000
HGB	20%	6,108	614	491.63	122.37	1.248918	0.249898	1.093008-1.338290
HGB	50%	15,264	1,393	1,228.58	164.42	1.133829	0.566952	1.083534-1.168554

Fig. 9: Period-Local Lift by Call Depth.



LR is strongest at tight call depths. At 5% its interval is wholly above 1.0, whereas HGB 5% crosses 1.0. The tested depths are sensitivity points, not prescribed quotas.

10. Final Model Selection and Pilot Interpretation

The final pilot specification is **unweighted Logistic Regression with all 12 valid pre-call predictors**. HGB remains a challenger. HGB is stronger on conventional shuffled evaluation, but LR is more robust in ordered evaluation and materially stronger at low call-depth lift.

Table 15: Why the pilot recommendation favours LR

Evidence	LR	HGB	Implication
Tuned holdout PR-AUC	0.240265	0.272820	HGB stronger conventional holdout
Shuffled ROC-AUC	0.684767	0.717461	HGB stronger shuffled
Ordered ROC-AUC	0.566290	0.551600	LR stronger future-like test
Ordered PR-AUC	0.112626	0.096333	LR stronger ordered ranking
5% lift	1.853857	1.284690	LR stronger at tight capacity
Interpretability/monitoring	Higher	Lower	LR easier to inspect and challenge

10.1 Controlled pilot

Use the same eligible pool and equal assigned call capacity for the LR-ranked arm and BAU/random comparator. Primary outcome: subscriptions per 1,000 assigned calls. Secondary controls: contactability, complaints, opt-outs, repeat contacts, customer mix and agent time where available. HGB may be shadow-scored or tested as a separately randomised challenger.

Decision gate

APPROVE for controlled pilot planning. DO NOT APPROVE for unrestricted production deployment until prospective incremental value, stability, calibration and economics are measured.

11. Limitations and Required Next Evidence

11.1 Temporal uncertainty

Month labels have no verified year/timestamp. Ordered blocks approximate source order rather than confirmed calendar deployment periods.

11.2 Customer identity

No true customer identifier exists. Pseudo-profile grouping cannot confirm repeated-customer independence.

11.3 Calibration

No prospective recalibration model is fitted. Raw scores are ranking signals only.

11.4 Historical association is not causal

Feature associations and importances can reflect campaign design, availability and customer composition. They do not establish causal effects.

11.5 Economics

The file lacks call cost, servicing/acquisition cost and net contribution per subscription. A defensible ROI cannot be estimated from this dataset alone. Pilot economics should use measured incremental subscriptions and real bank cost inputs.

11.6 Scope

The project deliberately compares a transparent linear model with one nonlinear challenger. Given weak ordered robustness, collecting better temporal/customer identifiers is more important than expanding the classifier catalogue.

12. Reproducibility and Evidence Traceability

The repository is intentionally flat. All five notebooks read term-deposit-marketing-2020-labelled.csv from repository root and do not depend on generated data/results/figures folders, caches or HTML exports. requirements.txt pins the execution environment.

Table 16: Notebook-to-report traceability

Notebook	Purpose	Report evidence
01_data_eda.ipynb	Audit, associations, ordered blocks	Tables 1-2; Figs. 1-2
02_baseline_tuning.ipynb	Leakage, tuning, holdout	Tables 3-5; Fig. 3
03_features_imbalance.ipynb	Importance, reduction, imbalance	Tables 6-9; Figs. 4-5
04_segments_robustness.ipynb	Population and robustness	Tables 10-12; Figs. 6-7
05_final_validation_reporting.ipynb	Calibration and lift	Tables 13-14; Figs. 8-9

Cross-notebook reconciliation confirms exact metric flow: Notebook 2 tuned pre-call metrics match Notebook 3 All-12; Notebook 4 ordered metrics match Notebook 5; the forward sample is consistently eight eligible periods, 30,526 rows and 2,457 positives. The final audit recorded 99 passes and zero failures.

12.1 Reproduction order

Install requirements.txt and execute 01 -> 02 -> 03 -> 04 -> 05. Notebook 2 performs tuning; Notebooks 3-5 reuse the selected pre-call structures for downstream tests.

Evidence standard

Every analytical table and figure in this report comes from the final executed notebooks. No confusion matrices, report-only predictive analyses, deployable duration features, sigmoid/isotonic calibration results or cached historical result files are used.

End of Technical Report